The LQD Model, from Distribution to Smile

An alternative lecture on log-quantile-density parametrization

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Abstract

A smile is a probability distribution wearing option-market clothes. The LQD model takes that sentence literally: instead of drawing implied volatility and then asking whether the resulting curve hides a negative density, it draws an increasing quantile function and prices the smile from the resulting law. For the log-forward return X, the model writes the logarithm of the quantile density as a universal two-sided tail skeleton plus a short Legendre expansion. Positivity is then automatic, a single additive shift enforces the forward martingale, and the endpoint scales identify both the last finite moments and the Lee wing slopes. This lecture develops the construction from a concrete logistic slice, proves the one-expiry no-butterfly result, derives pricing, tails, exact ATM level/skew/curvature identities, calibration, the analytic price Jacobian, and soft cross-expiry control. It then follows the production calibrator through two deterministic approximation cases: a seven-parameter fit to an SPX-like SVI target with maximum quote-grid error 0.23 volatility basis points, and a thirteen-parameter fit that qualitatively recovers a two-humped event density. The emphasis throughout is on the few facts that carry the model; numerical plumbing and control tables are kept out of their way.

Contents

1 The desk problem: do not manufacture negative probability

Imagine fitting a quiet expiry with seven liquid strikes. The quoted vols are unremarkable; the interpolated smile looks smooth; the residuals are tiny. Then someone asks for a tight butterfly between two quote nodes. The model returns a negative price. Nothing dramatic happened on the screen. The curve was simply smooth in the wrong coordinate, and its second strike derivative changed sign between observations.

That failure is easy to explain after the fact. A call curve must be decreasing and convex in strike, because its second strike derivative is a risk-neutral density. A generic interpolant in implied volatility does not know this. There are perfectly good direct-price and constrained-SVI approaches, but they must encode or police the same global condition. LQD makes a different trade: it starts from a probability law, where non-negative mass is local and cheap, then integrates to option prices.

Invariant

1. Every feasible one-expiry parameter vector defines a continuous, non-negative probability density and therefore a butterfly-free call slice.
2. The normalized underlying has mean one. This requires one scalar shift and, for the LQD tail family, the hard condition $A_R < 1$.
3. Tail behaviour is part of the distribution, not an extrapolation patch: the endpoint scales determine the critical moments and Lee slopes.
4. The trader handles — ATM volatility, log-strike skew and curvature — are analytic functionals of the constructed slice, not finite differences of an IV grid.

1.1 Normalize first, so carry does not clutter the mathematics

Fix an expiry date T. Let F_T and D_T be its forward and discount factor, and define

$$Y = \frac{S_T}{F_T}, \qquad X = \log Y, \qquad k = \log \frac{K}{F_T}. \tag{1}$$

Under deterministic carry and the T-forward measure,

$$\mathbb{E}[Y] = \mathbb{E}[e^X] = 1. \tag{2}$$

The normalized undiscounted call is

$$C(k) = \frac{C^{\$}(F_T e^k, T)}{D_T F_T} = \mathbb{E}\Big[(e^X - e^k)^+\Big]. \tag{3}$$

All slice formulas below live in these units. Dollars return only at the outer API boundary.

The two clocks. T labels an expiry date; τ denotes the year fraction used for variance annualization. In production, τ is the active variance time supplied as `prepared.tau`: it equals calendar time when the event clock of Note 11 is inactive, and otherwise includes scheduled-event dilation. Thus $w = \sigma^2 \tau$ and Black vega is $\varphi(d_+)\sqrt{\tau}$. Keeping τ visible avoids the common mistake of mixing carry time with variance time.

1.2 The model in one equation

Let $Q(u)$ be the quantile of X at percentile $u \in (0,1)$ and $q(u) = Q'(u)$ its quantile density. LQD stands for *log quantile density*: it models $\ell(u) = \log q(u)$ as

$$\boxed{\ell(u) = -\log u - \log(1-u) + (1-u)L + uR + \sum_{n=2}^{N} a_n P_n(1-2u)\ .} \tag{4}$$

Here P_n is the nth Legendre polynomial. The shipped order $N = 6$ has the seven coefficients $(L, R, a_2, \ldots, a_6)$; the API accepts $4 \le N \le 16$.

There are only three ideas in (4).

1. Taking a logarithm makes $q = e^{\ell}$ positive for every finite coefficient vector. Hence Q is increasing and cannot fold probability mass over itself.
2. The terms $-\log u - \log(1-u)$ impose controllable exponential tails on X, hence power tails on $Y = e^X$ and linear total-variance wings.
3. A short orthogonal-polynomial expansion bends the distribution between the endpoints.

One warning is worth placing beside the definition. L and R are convenient low-order endpoint *coordinates*; once $a_n \neq 0$, they are not the log tail scales. Every Legendre mode reaches both endpoints. The actual tail handles A_L, A_R will appear in section 5.

The full pipeline is

$$(L, R, a_2, \ldots, a_N) \longrightarrow q \longrightarrow Q \longrightarrow \text{mean-one shift} \longrightarrow C(k) \longrightarrow \sigma_{\text{imp}}(k).$$

In production, structural density positivity is still visible in one line:

Listing 1: The non-negative return density, distilled from `backend/volfit/models/lqd/quadrature.py`.

```
def density_at_quantile(u, g):
    # q(u) = exp(g(u)) / (u * (1-u))
    return u * (1-u) * np.exp(-g)  # f_X(Q(u)) = 1/q(u) >= 0
```

2 A probability ruler instead of a volatility curve

The cleanest way to read a quantile model is to start with a uniform random variable. Let $U \sim \text{Uniform}(0,1)$ and set

$$X = Q(U). \tag{5}$$

If Q is strictly increasing, this construction assigns exactly u units of probability below $Q(u)$. Where a density exists, differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u))Q'(u) = 1, \qquad f_X(Q(u)) = \frac{1}{q(u)}. \tag{6}$$

A large q means that a wide interval of returns is needed to accommodate a small interval of probability: the ordinary density is thin there. A small q means probability percentiles are packed together: the ordinary density is high. Quantile density and ordinary density are reciprocal views of the same ruler.

2.1 Why positivity of the ruler kills butterflies

Let $y = e^k$ be normalized strike and $\widetilde{C}(y) = \mathbb{E}[(Y - y)^+]$. When Y has a density,

$$\frac{\partial \widetilde{C}}{\partial y} = -(1 - F_Y(y)), \qquad \frac{\partial^2 \widetilde{C}}{\partial y^2} = f_Y(y) \geq 0. \tag{7}$$

This is the Breeden–Litzenberger identity. It is useful intuition, but the no-butterfly proof is even shorter.

Proposition 1 (Structural butterfly freedom)**.** *Suppose $\overline{Q}$ is strictly increasing and $M = \int_0^1 e^{\overline{Q}(u)}\mathrm{d}u$ is finite. Define $Q = \overline{Q} - \log M$. Then (3) is the call curve of a positive mean-one random variable. It is decreasing and convex in normalized strike, satisfies put–call parity, and is free of butterfly arbitrage.*

Proof. With U uniform, $Y = e^{Q(U)}$ is positive and $\mathbb{E}[Y] = e^{-\log M} M = 1$. A call is the expectation of the convex payoff $(Y - y)^+$. Its strike monotonicity and convexity follow either pointwise from the payoff or from (7); parity follows by subtracting the corresponding put payoff. No density-positivity constraint is left for the optimizer. □

The proposition is deliberately modest. LQD spans a finite-dimensional *subset* of continuous densities; it does not claim to represent every density. And the statement is for one expiry. Cross-expiry calendar arbitrage is a separate question (section 13). The scalar normalization translates the log-return law and rescales Y, but $Q' = \overline{Q}'$: quantile-density shape and tail slopes are unchanged.

2.2 Pricing directly on percentiles

Let u_k solve $Q(u_k) = k$. Only percentiles above u_k finish in the money, so

$$C(k) = \int_{u_k}^1 \left(e^{Q(u)} - e^k\right) \mathrm{d}u, \tag{8}$$
$$P(k) = \int_0^{u_k} \left(e^k - e^{Q(u)}\right) \mathrm{d}u. \tag{9}$$

Subtracting the two integrals and using $\int_0^1 e^{Q(u)}\mathrm{d}u = 1$ gives $C(k) - P(k) = 1 - e^k$.

3 The logistic slice: the whole machine in one toy

Before adding a single polynomial, set

$$a_n = 0, \qquad L = R = \log s, \qquad 0 < s < 1. \tag{10}$$

Then

$$q(u) = \frac{s}{u(1-u)}, \qquad Q(u) = \mu + s \log \frac{u}{1-u}.$$

So X is logistic with scale s, shifted by μ. The martingale integral is a beta integral:

$$\mathbb{E}[e^X] = e^\mu \int_0^1 u^s(1-u)^{-s}\mathrm{d}u = e^\mu B(1+s, 1-s) = e^\mu \frac{\pi s}{\sin(\pi s)}. \tag{11}$$

Consequently

$$\mu = -\log \frac{\pi s}{\sin(\pi s)}, \qquad s < 1, \tag{12}$$

and $\mathrm{Var}(X) = \pi^2 s^2/3$. The same inequality $s < 1$ that makes the beta integral finite will later become the general right-tail condition $A_R < 1$.

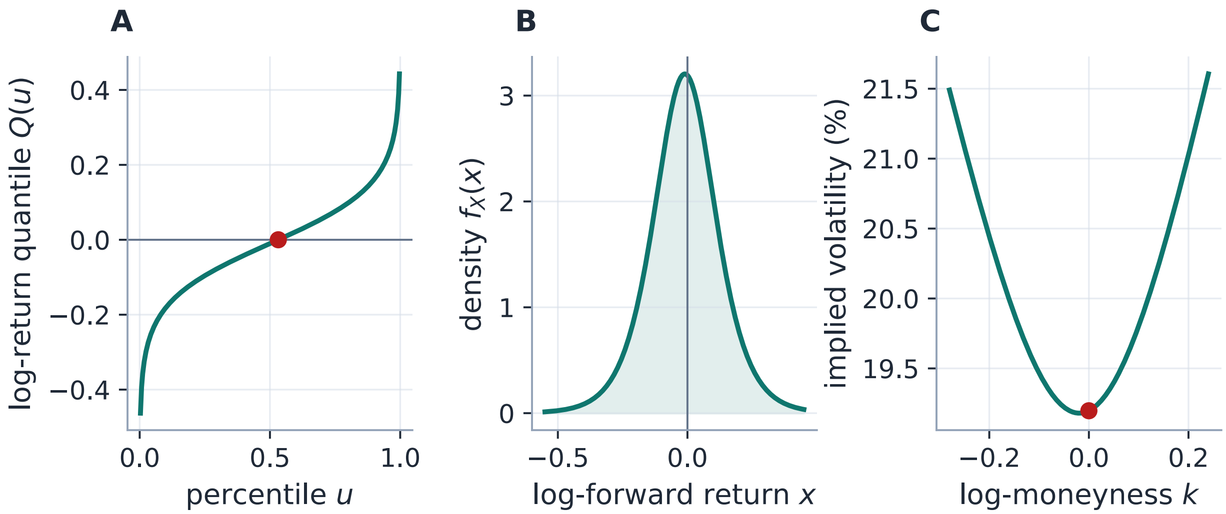

Figure 1: One law, three market languages. A logistic LQD slice with $s = 0.078$ becomes an increasing log-return quantile (A), a non-negative density (B), and a half-year implied-volatility smile (C). The red dot follows the forward strike: martingale normalization places $Q(u) = 0$ at $u = 0.5321$, not at the median, so a symmetric log-return shape does not imply a perfectly symmetric Black smile.

The plotted half-year slice has $\mu = -0.010028$, ATM volatility 19.20%, and a martingale self-check of $\mathbb{E}[e^X] - 1 = 0.00e+00$. Its two endpoint scales both equal 0.078, but the left and right Lee slopes differ: 0.0375 and 0.0406. The difference comes from the martingale moment on the right: a finite forward consumes one power of Y.

> **Example: why the cold start is close, but not exact**
>
> Production initializes s from the heuristic variance match $w_{\mathrm{ATM}} \approx \mathrm{Var}(X) = \pi^2 s^2/3$. This is intentionally cheap; ATM implied variance is not log-return variance. In fact, as $s \downarrow 0$,
>
> $$C(0) \sim s \log 2, \qquad B(0, w) \sim \sqrt{\frac{w}{2\pi}}, \qquad w_{\mathrm{ATM}} \sim 2\pi (\log 2)^2 s^2.$$
>
> The variance-match coefficient is about 8% larger. The initializer only needs to land in the right basin; the nonlinear fit supplies the correction.

The logistic law is therefore more than a classroom special case. It is a complete admissible slice, an analytic check on normalization and tails, and the actual cold-start family used by `logistic_init`.

4 Why the LQD formula has exactly this shape

4.1 The logit coordinate removes both endpoint singularities

Write the smooth part of (4) as

$$g(u) = (1-u)L + uR + \sum_{n=2}^{N} a_n P_n(1-2u). \tag{13}$$

The divergence of $q = e^g/[u(1-u)]$ is real: it creates the return tails. The logit coordinate absorbs that divergence into a finite tail slope. Set

$$z = \log\frac{u}{1-u}, \qquad u = \Lambda(z) = \frac{1}{1+e^{-z}}, \qquad \frac{\mathrm{d}u}{\mathrm{d}z} = u(1-u). \tag{14}$$

The two endpoint factors cancel:

$$\frac{\mathrm{d}Q}{\mathrm{d}z} = \frac{\mathrm{d}Q}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}z} = \frac{e^{g(u)}}{u(1-u)}u(1-u) = e^{g(\Lambda(z))} > 0. \tag{15}$$

This is the operative equation in code. The open percentile interval $(0,1)$ has become the real line, and the quantile is the integral of a smooth positive function:

$$\overline{Q}(z) = \int_0^z e^{g(\Lambda(r))}\mathrm{d}r. \tag{16}$$

Heuristic

Think of z as a probability ruler with evenly spaced tail marks. The function e^g is the local exchange rate between that ruler and log return. Large e^g stretches return space and thins the density; small e^g compresses return space and piles probability up. Positivity says the ruler may stretch or compress, but it may never run backwards.

4.2 Why Legendre polynomials, and why start at degree two?

With $x = 1-2u$, Legendre polynomials obey

$$\int_0^1 P_m(1-2u)P_n(1-2u)\mathrm{d}u = \frac{\mathbf{1}_{m=n}}{2n+1}. \tag{17}$$

They provide a numerically well-scaled global basis on the percentile interval. The constant and linear degrees are already carried by

$$(1-u)L + uR = \frac{L+R}{2} + \frac{L-R}{2}(1-2u), \tag{18}$$

so the explicit sum begins at $n = 2$. At $N = 6$ the model spans the same polynomial space as $P_0, \ldots, P_6$, while retaining two readable baseline endpoint coordinates.

The word *body mode* can mislead. Legendre polynomials are global: $P_n(1) = 1$ and $P_n(-1) = (-1)^n$. A coefficient that sculpts the centre also moves one or both endpoint scales. Orthogonality improves conditioning; it does not localize the parameters. Production evaluates the basis by the three-term recursion

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \qquad P_0 = 1, \quad P_1 = x. \tag{19}$$

5 The wings know their moments

5.1 Endpoint scales are the genuine tail handles

Because g is continuous at both endpoints,

$$A_L := e^{g(0)} = \exp\left(L + \sum_{n=2}^{N} a_n\right), \tag{20}$$

$$A_R := e^{g(1)} = \exp\left(R + \sum_{n=2}^{N} (-1)^n a_n\right). \tag{21}$$

Integrating $q(u) \sim A_L/u$ on the left and $q(u) \sim A_R/(1-u)$ on the right gives finite constants B_L, B_R such that

$$Q(u) = B_L + A_L \log u + o(1) \quad (u \downarrow 0), \qquad Q(u) = B_R - A_R \log(1-u) + o(1) \quad (u \uparrow 1). \tag{22}$$

Thus X has exponential tails. Equivalently, $Y = e^X$ has power tails. This is a modelling assumption, not a mere numerical convenience.

5.2 One integral gives the moment strip and the hard wall

For any real r,

$$\mathbb{E}[e^{rX}] = \int_0^1 e^{rQ(u)} \mathrm{d}u.$$

Using (22), the left integrand behaves like u^{rA_L} and the right like $(1-u)^{-rA_R}$. Therefore

$$\mathbb{E}[e^{rX}] < \infty \quad \Longleftrightarrow \quad -\frac{1}{A_L} < r < \frac{1}{A_R}. \tag{23}$$

Setting $r = 1$ proves the load-bearing admissibility condition

$$A_R < 1. \tag{24}$$

For finite coefficients this condition is necessary and sufficient for the forward moment. The left tail needs no analogous restriction because $A_L > 0$ already makes $r = 1$ integrable there.

> **Caution**
>
> $A_R = 1$ is not a regularization preference. At that wall the normalized underlying has infinite mean, so no finite forward exists. Production keeps a 10^{-6} buffer, rejects $A_R \geq 1 - 10^{-6}$, and adds a soft barrier centred at 0.90 to turn the optimizer back before rejection. The barrier is

smooth inside the feasible region; the switch to the infeasible penalty branch is not an algebraic continuation through the wall.

5.3 From critical moments to Lee slopes

For $Y = e^X$, define the last finite extra right moment and inverse left moment

$$p_+ = \sup\{p \geq 0 : \mathbb{E}[Y^{1+p}] < \infty\} = \frac{1}{A_R} - 1, \qquad p_- = \sup\{p \geq 0 : \mathbb{E}[Y^{-p}] < \infty\} = \frac{1}{A_L}. \tag{25}$$

Lee's moment map is

$$\psi(p) = 2 - 4\left(\sqrt{p^2 + p} - p\right), \qquad p \geq 0. \tag{26}$$

Under the regular exponential-tail asymptotics of (22), the total implied variance $w(k)$ has the limiting slopes

$$\beta_L := \lim_{k \to -\infty} \frac{w(k)}{|k|} = \psi(p_-), \qquad \beta_R := \lim_{k \to +\infty} \frac{w(k)}{k} = \psi(p_+). \tag{27}$$

For a trader who wants to skip the critical-moment notation, the same maps are

$$\beta_L = 2\,\frac{\sqrt{1 + A_L} - 1}{\sqrt{1 + A_L} + 1}, \qquad \beta_R = 2\,\frac{1 - \sqrt{1 - A_R}}{1 + \sqrt{1 - A_R}}, \quad 0 < A_R < 1. \tag{28}$$

Larger endpoint scale means a heavier return tail and a steeper variance wing. The right-hand curve reaches the model-free ceiling 2 as the finite-forward wall is approached.

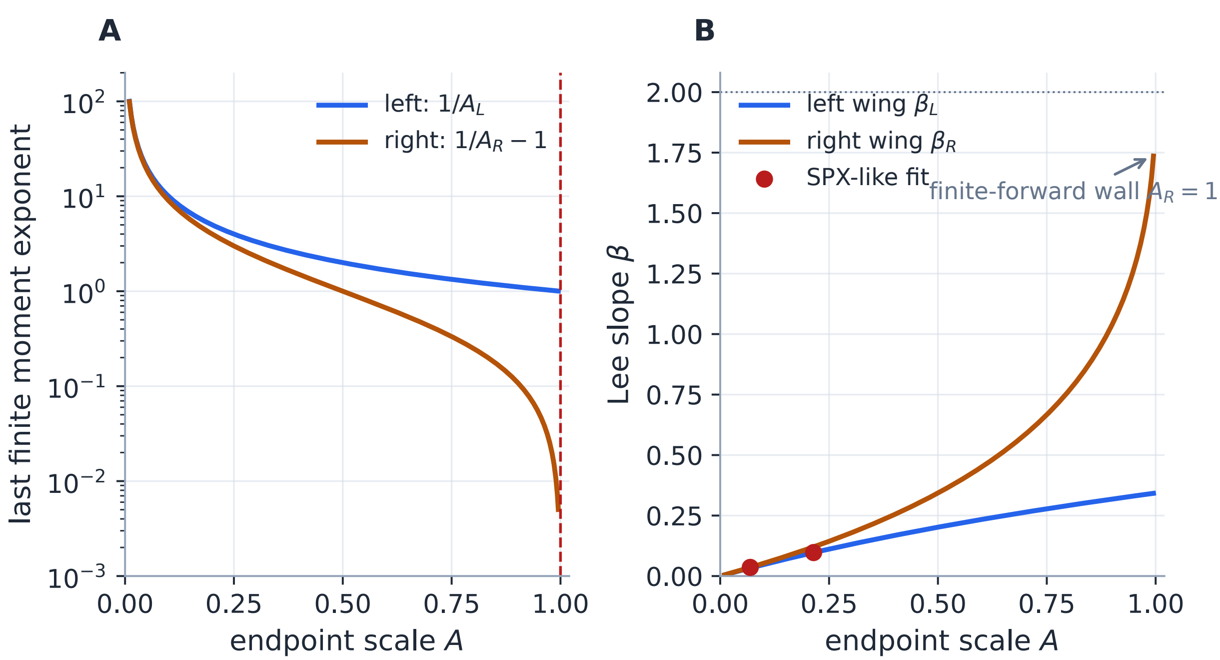

Figure 2: The endpoint scales carry economic meaning. Panel A maps each scale to its last finite moment; the right moment collapses at $A_R = 1$. Panel B maps the same scales to Lee slopes. The red points are the two tails of the SPX-like benchmark: they sit far from the wall and well below the slope ceiling.

Example: reading the SPX-like tails

The fitted benchmark later studied in section 10 has $A_L = 0.214458$ and $A_R = 0.069023$. Equations (25)–(27) give $\beta_L = 0.097073$ and $\beta_R = 0.035757$. These are sensible extrapolation diagnostics, not strongly identified market observables: twenty-five quotes over a finite strike window cannot uniquely determine asymptotic moments.

6 Pricing is one cumulative integral

The percentile formula (8) is conceptually complete. Production re-expresses it on the logit grid because that grid is smooth, symmetric and shared by every strike.

6.1 Normalize once, then reuse the upper asset share

Start from the centred primitive (16). Since $\mathrm{d}u = u(1-u)\mathrm{d}z$, the martingale shift is

$$\mu = -\log\left[\int_{-\infty}^{\infty} e^{\overline{Q}(z)}\Lambda(z)(1-\Lambda(z))\mathrm{d}z\right], \qquad Q(z) = \mu + \overline{Q}(z). \tag{29}$$

Define the upper asset-share integral

$$\mathcal{A}(z) = \int_z^{\infty} e^{Q(r)}\Lambda(r)(1-\Lambda(r))\mathrm{d}r. \tag{30}$$

If z_k solves $Q(z_k) = k$ and $u_k = \Lambda(z_k)$, then

$$C(k) = \mathcal{A}(z_k) - e^k(1-u_k). \tag{31}$$

This is just (8) split into its asset and strike legs. One forward cumulative integral builds Q; one reverse cumulative integral builds $\mathcal{A}$; each strike then needs only a scalar inversion and interpolation.

6.2 From price to the quoted Black volatility

The normalized Black call in total-variance coordinates is

$$B(k,w) = \Phi(d_+) - e^k\Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}. \tag{32}$$

For an admissible call price, implied total variance solves $B(k, w(k)) = C(k)$, and

$$\sigma_{\mathrm{imp}}(k) = \sqrt{\frac{w(k)}{\tau}}. \tag{33}$$

LQD therefore does not interpolate in implied-volatility space. The full strike curve, including its extrapolation, is the Black image of a fitted probability distribution.

6.3 What the finite grid actually computes

The production grid is $z \in [-40, 40]$. Optimization uses 2001 nodes; the accepted slice is rebuilt once on 8001 nodes for reporting and downstream diagnostics. Cumulative Simpson integration constructs Q and $\mathcal{A}$ when available, while the martingale body integral uses the trapezoidal rule. At the two truncated endpoints the leading asymptotic corrections are

$$\int_{40}^{\infty} e^{Q}u(1-u)\mathrm{d}z \approx \frac{e^{Q(40)-40}}{1-A_R}, \qquad \int_{-\infty}^{-40} e^{Q}u(1-u)\mathrm{d}z \approx \frac{e^{Q(-40)-40}}{1+A_L}. \tag{34}$$

For a fixed finite-order polynomial g, the neglected relative terms decay exponentially in the cutoff. They are tiny for ordinary fitted equity scales, but (34) remains an asymptotic correction, not an exact tail integral.

At each grid node the implementation stores

$$Q_z = e^{g}, \qquad \mathcal{A}_z = -e^{Q}u(1-u).$$

These exact nodal derivatives define cubic-Hermite interpolants. Strike inversion starts from a linear seed and takes four safeguarded Newton steps. On benchmark slices this is much more accurate than a table lookup; it should not be advertised as a universal extrapolator outside the finite grid or as an algebraic proof of machine precision for arbitrary coefficients.

Performance

The arrays $(z, u, u(1-u))$ and the Legendre matrix depend only on grid size and model order. They are cached and returned read-only. Calibration reforms only g, its exponential and the cumulative integrals. The static-grid cache made `build_slice` about 1.7× faster; the two-grid scheme spends full resolution only on the accepted parameters.

7 Exact ATM handles: level, digital mismatch and density

Traders rarely want to drag a_4 on a screen. They want ATM volatility, ATM skew and ATM curvature. LQD obtains these as analytic log-strike derivatives of the numerically constructed slice, without finite-differencing an implied-vol grid.

Let u_0 solve $Q(u_0) = 0$ and define

$$c_0 = C(0), \qquad f_0 = f_X(0) = u_0(1-u_0)e^{-g(u_0)}. \tag{35}$$

Differentiating the call expectation with respect to log strike gives

$$C'(0) = -(1-u_0), \qquad C''(0) = f_0 - (1-u_0). \tag{36}$$

The ATM Black identity is

$$c_0 = 2\Phi\left(\frac{\sqrt{w_0}}{2}\right) - 1,$$

so its solution is explicit:

$$w_0 = 4\left[\Phi^{-1}\left(\frac{1+c_0}{2}\right)\right]^2. \tag{37}$$

Set

$$d_0 = \frac{\sqrt{w_0}}{2}, \qquad \varphi_0 = \varphi(d_0), \qquad \delta = u_0 - \Phi(d_0). \tag{38}$$

Implicit differentiation of $B(k, w(k)) = C(k)$ and elementary simplification give

$$w'(0) = 2\sqrt{w_0}\,\frac{\delta}{\varphi_0}, \tag{39}$$

$$w''(0) = 2\sqrt{w_0}\,\frac{f_0}{\varphi_0} - 2 + 2\left(1 + \frac{w_0}{4}\right)\left(\frac{\delta}{\varphi_0}\right)^2. \tag{40}$$

Converting total variance to annualized volatility yields the three handles

$$\sigma_0 = \sqrt{\frac{w_0}{\tau}}, \tag{41}$$

$$s_0 = \left.\frac{\partial \sigma_{\mathrm{imp}}}{\partial k}\right|_0 = \frac{\delta}{\varphi_0\sqrt{\tau}}, \tag{42}$$

$$\kappa_0 = \left.\frac{\partial^2 \sigma_{\mathrm{imp}}}{\partial k^2}\right|_0 = \frac{1}{\sqrt{\tau}}\left[\frac{f_0}{\varphi_0} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4}\left(\frac{\delta}{\varphi_0}\right)^2\right]. \tag{43}$$

Proposition 2 (What the ATM handles measure)**.** *For any admissible numerically constructed LQD slice, the identities (41)–(43) give its exact analytic log-strike ATM derivatives. Level is determined by the ATM call, skew by the difference between the LQD and Black ATM CDF levels u_0 and $\Phi(d_0)$ (equivalently, the negative call-digital mismatch), and curvature primarily by the density height f_0, with a skew correction.*

Proof. Equation (36) follows from $C'(k) = -e^k(1 - F_X(k))$. Differentiate the identity $B(k, w(k)) = C(k)$ once and twice, substitute the Black partial derivatives at $k = 0$, and simplify using (38). Finally differentiate $\sigma_{\mathrm{imp}} = \sqrt{w/\tau}$. □

The qualifier *numerically constructed* matters. u_0, c_0 and f_0 come from the quadrature slice. The formulas are exact chain-rule identities conditional on that slice; they are not closed-form functions of the raw coefficient vector.

8 A local chart in trader coordinates

Let $\theta = (L, R, a_2, \ldots, a_N) \in \mathbb{R}^d$ and collect the internal chart handles as

$$H(\theta) = (w_0, s_0, \kappa_0). \tag{44}$$

The graph-facing carrier used by Note 14 replaces w_0 with $\sigma_0 = \sqrt{w_0/\tau}$; the distinction is small in notation and important in units.

Assumption 1 (Regular handle point)**.** At the reference slice θ^*, the Jacobian $J = \partial H/\partial\theta \in \mathbb{R}^{3\times d}$ has full row rank three and is not numerically ill-conditioned in the chosen Euclidean coefficient metric.

Under assumption 1, let $J^\dagger = J^\mathsf{T}(JJ^\mathsf{T})^{-1}$ be the least-norm right inverse and let the columns of V form an orthonormal basis of $\ker J$. A local displacement can be written

$$\theta = \theta^* + J^\dagger\alpha + V\xi. \tag{45}$$

To first order, $\alpha \in \mathbb{R}^3$ moves the three trader handles one-for-one, while the shape coordinate $\xi \in \mathbb{R}^{d-3}$ leaves them unchanged: $JJ^\dagger = I_3$ and $JV = 0$. In code, an SVD/QR construction is preferable near weak rank because it exposes the small singular values rather than hiding them in $(JJ^\mathsf{T})^{-1}$.

For a finite move, first-order orthogonality is not enough. Production holds ξ fixed and solves the three-dimensional nonlinear equation

$$H(\theta^* + J^\dagger \alpha + V\xi) = H_{\text{target}} \tag{46}$$

by Newton iteration. This keeps the chart's local shape coordinates fixed; it does *not* promise that the endpoint scales or wings remain unchanged.

Caution

The chart is local and conditional on rank. A large handle request can leave the neighbourhood where J is useful, drive A_R toward the integrability wall, or admit no feasible retargeted slice. The chart separates tasks; it does not regularize the free shape coordinates. Ridge penalties, data and priors remain responsible for suppressing implausible oscillation.

9 Calibration: fit prices, measure errors in volatility units

9.1 The core objective

Suppose the prepared quote set contains $(k_i, \sigma_i, \omega_i)$ at active time τ, with quote total variance $w_i = \sigma_i^2 \tau$. Convert each quote once to its normalized Black call and freeze its quote vega:

$$C_i^{\text{mkt}} = B(k_i, w_i), \qquad v_i = \varphi(d_{+,i})\sqrt{\tau}. \tag{47}$$

The core fit minimizes

$$\sum_i \omega_i \left(\frac{C^{\text{LQD}}(k_i;\theta) - C_i^{\text{mkt}}}{v_i + \eta} \right)^2 + \lambda \sum_{n=4}^{N} n^{2r} a_n^2. \tag{48}$$

The price numerator preserves the distribution-first construction at each feasible priced iterate. Dividing by quote vega makes the residual locally comparable to a volatility error; the floor $\eta = 10^{-4}$ prevents far-wing quotes from receiving explosive weight. The high-order ridge leaves a_2, a_3 free and damps modes $n \geq 4$.

The low-level function defaults to $\lambda = 0$. The shipped application setting is $\lambda = 10^{-6}$, and the benchmark generator passes it explicitly. This distinction matters when reproducing a fit from the library rather than the UI.

9.2 The soft turn before the hard wall

The residual stack always contains the softplus row

$$b(\theta) = \log(1 + \exp\{s_b(A_R - c)\}), \qquad c = 0.90, \quad s_b = 50.0. \tag{49}$$

It is evaluated with a stable `logaddexp`. When a trial reaches the buffered hard wall $A_R \geq 1 - 10^{-6}$, slice construction is skipped and the price block is replaced by a large penalty increasing with A_R.

Thus an infeasible trust-region trial does not produce a fake option surface; it receives a direction back toward feasibility.

9.3 Initialization and warm starts

The cold start is the logistic family of section 3:

$$s_{\text{init}} = \frac{\sqrt{3w_{\text{ATM}}}}{\pi}, \qquad L = R = \log s_{\text{init}}, \qquad a_n = 0. \tag{50}$$

As the logistic example showed, this is a scale heuristic rather than an exact ATM match. In a surface sweep, the fitted shorter expiry is usually a much better initializer for the next expiry. The production workflow therefore fits nearest to farthest and warm-starts sequentially. A data-only prepass can also supply the measurement stage for the persistence and filtering machinery of Notes 13 and 15.

9.4 The production residual stack

The core objective is only the spine. `calibrate_slice` concatenates the blocks in table 1. Optional blocks are absent when inactive, and the Jacobian route changes with them.

Block	Activation	Scale	Jacobian
Mid or band quote data	selected fit mode	vega-normalized price	analytic
High-order ridge	$\lambda > 0$	coefficient residual	analytic
Calendar hinge	nearer fitted expiry available and control on	asset share	analytic
A_R barrier	always	dimensionless	analytic
Market var-swap	active quote and toggle	volatility	FD
Strike-gap prior	active prior mode	vega-normalized price	FD
Prior var-swap	active companion prior	volatility	FD
Operator/filter prior	gated prior or active filter	handle/operator units	FD

Table 1: The production residual stack. FD denotes scipy's two-point finite-difference Jacobian; activating any var-swap or prior block switches the entire solve to that route.

10 Case file: an SPX-like smile

The first test is deliberately synthetic. It asks whether seven LQD parameters can approximate a clean, familiar equity-index shape; it is not a claim about live market fit. At $\tau = 0.5$, the target is raw SVI

$$w_{\text{SVI}}(k) = a + b\left\{\rho(k-m) + \sqrt{(k-m)^2 + \sigma_{\text{SVI}}^2}\right\}, \tag{51}$$

with

$$(a, b, \rho, m, \sigma_{\text{SVI}}) = (0.010625,\ 0.07289,\ -0.5,\ 0.05831,\ 0.10100).$$

Twenty-five synthetic quotes cover $k \in [-0.35, 0.30]$. The production calibrator uses $N = 6$, the application ridge $\lambda = 10^{-6}$, equal quote weights, the logistic cold start and the standard barrier.

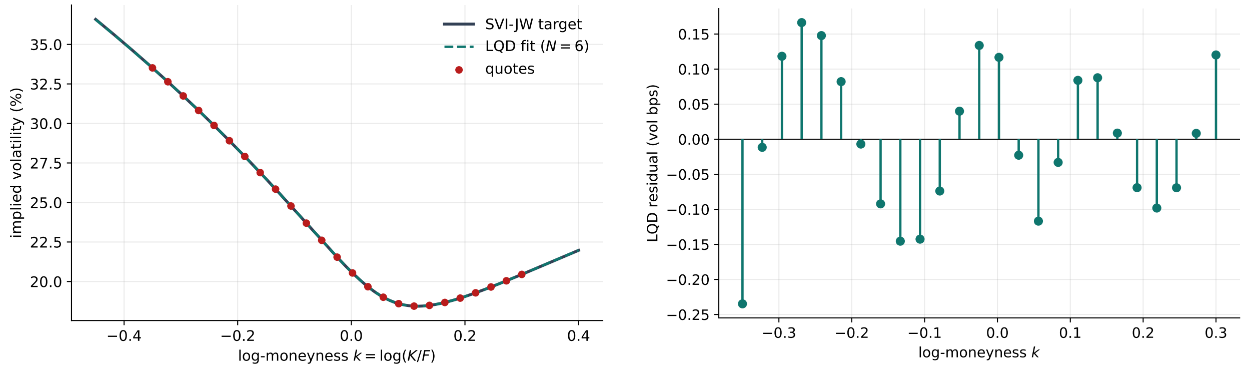

Figure 3: The seven-parameter LQD slice is visually indistinguishable from the SVI target over the quote window (left), but the residual panel (right) makes the approximation claim auditable: the maximum quote-grid error is 0.23 volatility basis points. Outside the fitted window the LQD curve follows its own distributional tails, not an SVI continuation.

The fit takes 7 residual evaluations. Its exact ATM readout is

$$\sigma_0 = 20.62\%, \qquad s_0 = -0.3538, \qquad \kappa_0 = 1.6474.$$

The martingale self-consistency check is $\mathbb{E}[e^X] - 1 = 0.00e+00$. The endpoint and tail diagnostics are

$$A_L = 0.214458, \quad A_R = 0.069023, \quad \beta_L = 0.097073, \quad \beta_R = 0.035757.$$

Raw SVI has asymptotic slopes $b(1-\rho) = 0.1093$ on the left and $b(1+\rho) = 0.0364$ on the right. Their closeness to the LQD diagnostics is reassuring for this constructed target, but finite-window quotes do not in general identify asymptotic slopes to two decimals.

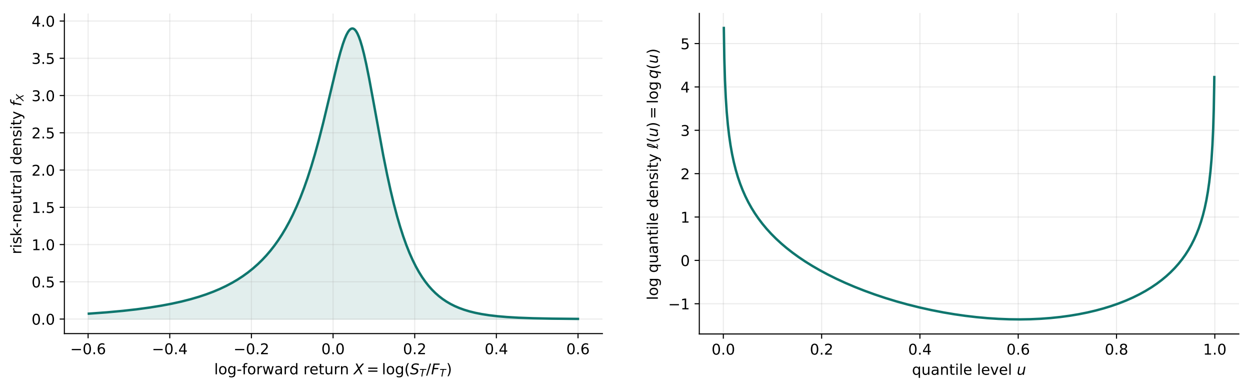

Figure 4: The same fit in distribution coordinates. The log-return density is non-negative because it is the reciprocal of a positive quantile density (left). The log quantile density (right) shows the endpoint rise that produces exponential return tails. Positivity is structural; the economic plausibility of the shape still comes from data and regularization.

Table 2: Fresh seven-parameter LQD fit to the SPX-like SVI target. Because the basis is global, the coefficients should not be read individually as pure level, skew, curvature or wing knobs.

Parameter	Value
L	-1.87479110
R	-3.08957252
a_2	+0.38076853
a_3	-0.04700855
a_4	-0.00719617
a_5	+0.00645578
a_6	+0.00213226

Case verdict: approximation capacity, not market alpha

Setup: a clean six-month index-like SVI smile. **Failure to avoid:** matching the quotes while creating negative density between them. **Mechanism:** fit normalized prices inside the LQD density family. **Verdict:** 0.23 vol bp maximum error on the twenty-five input quotes, a non-negative density and a finite-forward tail far from the wall. Live bid–ask and out-of-sample comparisons belong to `backend/backtest/`, not to this synthetic case.

11 Case file: the density wears two hats

Scheduled elections, court decisions, earnings and binary regulatory events can produce a terminal law with two plausible landing zones. To isolate that geometry, take the equal mixture

$$X \sim \frac{1}{2}N(m_1, s^2) + \frac{1}{2}N(m_2, s^2), \qquad (m_1, m_2, s) = (-0.10075573,\ 0.08924427,\ 0.05). \tag{52}$$

Here $s = 5\%$ is the component standard deviation of the thirty-day *log-return*, not an annualized implied volatility. The means are chosen so

$$\frac{1}{2}e^{m_1+s^2/2} + \frac{1}{2}e^{m_2+s^2/2} = 1, \tag{53}$$

which martingale-normalizes the mixture. Its call is available in closed form:

$$C_{\text{mix}}(k) = \frac{1}{2}\sum_{j=1}^{2}\left[e^{m_j+s^2/2}\Phi\left(\frac{m_j+s^2-k}{s}\right) - e^k\Phi\left(\frac{m_j-k}{s}\right)\right]. \tag{54}$$

We turn forty-one such calls over $k \in [-0.25, 0.25]$ into implied variances and fit $N = 12$ with $\lambda = 10^{-7}$.

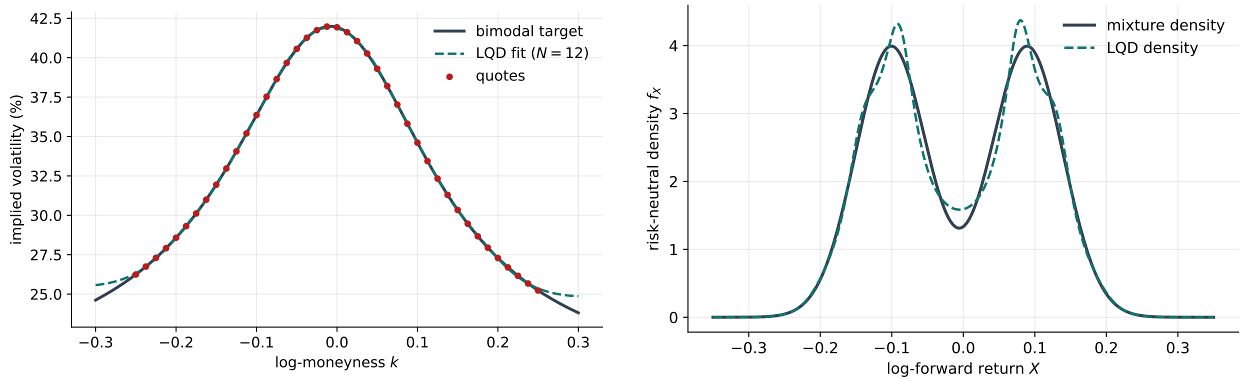

Figure 5: A binary-event geometry needs more than a smooth U-shaped smile. The thirteen-parameter fit captures the central trough and supported wings (left) and qualitatively recovers both modes of the known target density (right). The maximum quote-grid error is 13.19 vol bp; the density comparison is possible only because this is a synthetic case with a known law.

The fitted endpoint scales are $A_L = 0.015341$ and $A_R = 0.014979$, both thin and far from the integrability wall. Higher order is doing real work in the centre, but positivity alone does not certify that the two modes are genuine. With sparse quotes, a high-order LQD slice can create a beautifully admissible fiction.

Case verdict: positive is not the same as plausible

Setup: a known two-normal event mixture. **Failure to avoid:** forcing a low-order smile through a central trough, or using an unconstrained flexible curve that turns negative between quotes. **Mechanism:** higher Legendre order inside an always-positive quantile density. **Verdict:** both modes are qualitatively recovered, with 13.19 vol bp maximum quote-grid error. On real data, order selection, ridge strength, bid–ask information and event priors must decide whether a second hat is signal or overfit.

12 The free derivative: an envelope argument

For a seven-parameter fit, a finite-difference Jacobian rebuilds the quadrature once for the value and once per bumped coefficient. That is tolerable; at higher order and across many expiries it becomes the dominant repeated work. The continuous price formula contains a fortunate cancellation.

Let $u_k(\theta)$ be the moving percentile satisfying $Q(u_k(\theta);\theta) = k$. Differentiate (8) at fixed strike:

$$\begin{aligned}\frac{\partial C(k;\theta)}{\partial \theta} &= \int_{u_k}^{1} e^{Q(u;\theta)} Q_\theta(u;\theta)\mathrm{d}u - \left[e^{Q(u_k;\theta)} - e^{k}\right]\frac{\partial u_k}{\partial \theta} \\ &= \int_{u_k}^{1} e^{Q(u;\theta)} Q_\theta(u;\theta)\mathrm{d}u. \end{aligned} \tag{55}$$

The boundary term vanishes because $Q(u_k;\theta) = k$. The strike root may move violently with a parameter; its first-order effect on the payoff is still zero because the payoff itself is zero at the exercise boundary.

Theorem 1 (Implicit-root cancellation)**.** *For an admissible smooth LQD slice, the parameter derivative of a fixed-strike call is the explicit quantile sensitivity in* (55)*. No derivative of the implicit* *strike root is required.*

Proof. Apply Leibniz' rule to (8). The moving-boundary term is the payoff at the boundary times $u_{k,\theta}$, and the payoff is zero. $\square$

In logit coordinates, production evaluates the same derivative by interpolating the parameter sensitivities of the upper asset share at the fixed z_k. If $\phi_j(u) = \partial g/\partial\theta_j$, then

$$\frac{\partial \overline{Q}(z)}{\partial \theta_j} = \int_0^z e^{g(\Lambda(r))}\phi_j(\Lambda(r))\mathrm{d}r, \tag{56}$$

where

$$\phi_L = 1-u, \qquad \phi_R = u, \qquad \phi_{a_n} = P_n(1-2u).$$

Differentiate the martingale normalization, add the resulting scalar μ_{θ_j} to (56), and reverse-integrate $e^Q u(1-u)Q_{\theta_j}$ to obtain the asset-share sensitivity. The endpoint corrections also differentiate through $Q(\pm 40)$ and A_L, A_R. Ridge, calendar and barrier rows then add elementary derivatives.

The theorem is exact for the continuous integrals. The production price and gradient use separately constructed Hermite interpolants, so an off-node comparison inherits a small discretization mismatch. The tested claim is numerical: on mid, band and calendar residual stacks the analytic Jacobian agrees with a three-point finite difference to relative tolerance 10^{-3}, and analytic and two-point-FD fits reach coefficient vectors within about 10^{-5} and costs agreeing near 10^{-11} relative.

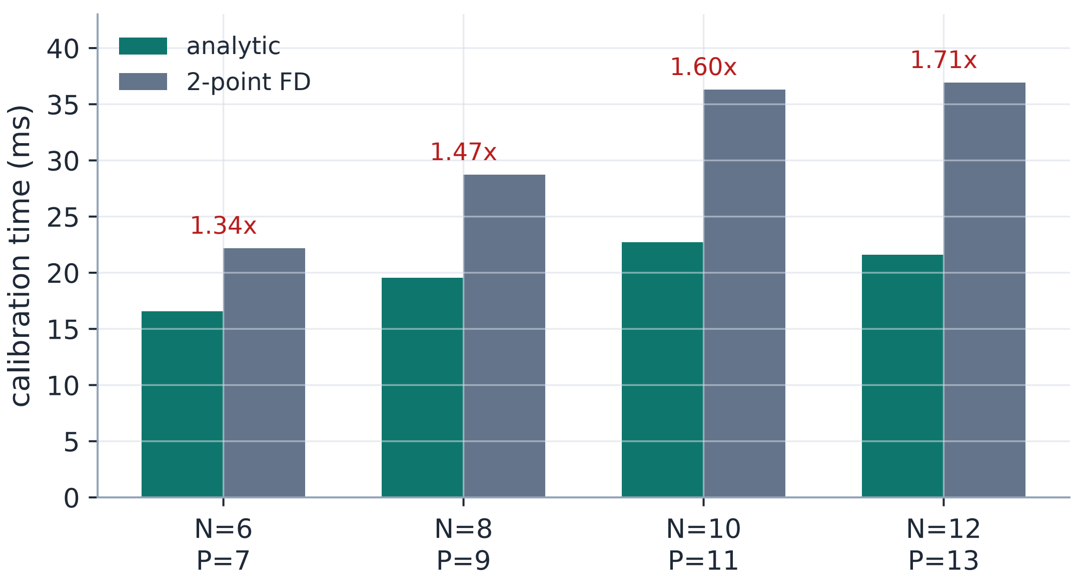

Figure 6: The envelope cancellation removes repeated quadrature builds. On the warm-cache optimizer-core benchmark, the analytic route is 1.34–1.71× faster over $N = 6, \ldots, 12$ and reaches the same optimum within the tested tolerances. These bars exclude the final full-grid rebuild and reporting work, so they are not end-to-end API latency.

Heuristic

Why is the speed-up not P-fold for P parameters? Cached static arrays make each bumped build relatively cheap, the trust-region solve still performs linear algebra and value evaluations, and these clean synthetic fits converge in few iterations. The useful result is structural: higher order no longer forces one full quadrature rebuild per Jacobian column.

13 From slices to a surface: convex order, softly enforced

Butterfly freedom is a within-expiry property. For expiry dates $T_1 < T_2$, absence of calendar arbitrage under deterministic carry requires

$$C_{T_1}(k) \le C_{T_2}(k) \qquad \text{for every } k. \tag{57}$$

Because both normalized underlyings have mean one, this is equivalent to saying that Y_{T_2} dominates Y_{T_1} in convex order: the later law is more dispersed without moving its mean.

Quantiles give a native criterion. Define the integrated upper quantile

$$G_T(\alpha) = \int_\alpha^1 e^{Q_T(u)} \mathrm{d}u, \qquad 0 \le \alpha \le 1. \tag{58}$$

For equal-mean positive laws,

$$Y_{T_1} \le_{cx} Y_{T_2} \quad \Longleftrightarrow \quad G_{T_1}(\alpha) \le G_{T_2}(\alpha) \quad \text{for every } \alpha, \tag{59}$$

with equality at $\alpha = 0$. In LQD, $G_T(\Lambda(z))$ is exactly the asset share $\mathcal{A}_T(z)$ already present on the pricing grid.

Production fits nearest to farthest. When calendar control is on, a new slice receives hinge residuals whenever its asset share falls below the previous slice at sampled logit nodes. The comparison is expressed at z values, so the same locations can be evaluated on either the optimization or reporting grid.

Caution

The mathematical condition (59) is a continuum condition. Production samples it, assigns a finite penalty weight, and allows the control to be switched off. It is therefore calendar *control*, not a hard guarantee. A later independent single-expiry refit has no neighbouring-slice context and can also break a previously coupled surface. Diagnostics must measure the final call ordering rather than infer it from the workflow.

Note 10 treats the model-agnostic calendar problem in detail. LQD's advantage is not a stronger theorem; it is that the natural convex-order object is already one of its cumulative pricing arrays.

14 The native log-contract route

For a mean-one positive underlying, the continuous log-contract proxy to total variance is

$$w_{\mathrm{vs}} = -2\mathbb{E}[X] = -2\int_0^1 Q(u)\mathrm{d}u = -2\int_{-\infty}^{\infty} Q(z)\Lambda(z)(1-\Lambda(z))\mathrm{d}z. \tag{60}$$

The last integral uses the grid LQD already owns. Production evaluates it by a trapezoidal rule without explicit tail corrections; its integrand decays as $ze^{-|z|}$, so the omitted $|z| > 40$ contribution is negligible in ordinary double precision.

Equation (60) prices a log contract. Identifying it with a realized-variance swap assumes the usual continuous-monitoring, continuous-path convention. Jumps and discrete sampling require corrections. Activating the market var-swap residual also switches the whole calibration to the finite-difference Jacobian because this optional block has not yet been added to the analytic route.

15 What is guaranteed, and what is merely hoped for

Table 3 is the shortest honest description of the model.

Property	Status	Qualification
Non-negative continuous density	Structural	For every finite coefficient vector; numerical pricing is still finite precision.
One-expiry butterfly freedom	Structural after normalization	Requires a finite forward and correct normalized units.
Finite forward	If and only if $A_R < 1$ mathematically	Code uses the buffer $A_R < 1 - 10^{-6}$.
Exponential X tails and Lee-consistent wings	Structural	A modelling choice; finite strikes weakly identify the scales.
ATM level, skew and curvature	Analytic slice identities	Conditional on the numerically built slice; not raw-coefficient closed forms.
Calendar freedom	Not structural	Sampled, finite-weight, optional control.
Economically plausible density	Not guaranteed	High order can fit noise or invent modes while remaining positive.
Atoms at zero or elsewhere	Not represented	The law is continuous; sharp continuous features only approximate mass points.
Unique or stable coefficients	Not guaranteed	Sparse quotes create nearly flat parameter directions; compare prices, handles and densities instead.
Market superiority	Not established here	The cases are deterministic synthetic approximation tests.

Table 3: The LQD contract. Arbitrage structure and economic plausibility are different promises.

Three further limitations matter in practice.

- **Near-wall conditioning.** As $A_R \uparrow 1$, the martingale tail correction and its derivatives dominate. A technically feasible slice close to the wall is fragile, and the orthogonal retarget can fail.
- **Global modes.** A local-looking smile improvement can alter both endpoint scales because every Legendre polynomial reaches the endpoints. Fitted coefficient vectors are therefore poor objects for cross-date comparison.
- **Density inference.** Option prices smooth the density twice. Even a sub-vol-bp fit does not identify narrow density features; digitals, one-touches and event probabilities deserve separate stability tests.

16 What the implementation exploits

Quantile models, density-first option pricing, Legendre expansions and Lee's moment formula are classical ingredients. The useful contribution is their combination into a production coordinate system:

1. The endpoint skeleton in (4) turns positivity and two-sided tail regularity into a short unconstrained expansion, with only the genuine right-moment wall left hard.
2. The ATM identities in section 7 translate the distribution into actual trader handles, while the local null-space chart separates handle moves from remaining first-order shape directions.
3. The envelope cancellation in theorem 1 lets the price Jacobian reuse the same cumulative integrals as pricing rather than differentiating an implicit strike root or rebuilding the slice per parameter.

None of these statements requires claiming that a generic calculus identity is new. The engineering value lies in choosing coordinates that make the identities cheap, testable and composable with the rest of the surface system.

17 Traceability

The figures and displayed numbers are deterministic outputs of `Docs/notes/figures/gen_lqd.py` and `Docs/notes/figures/gen_lqd_lecture.py`; LaTeX does not regenerate them automatically. Fresh generator values are the evidence used in this lecture. Some legacy benchmark fixtures intentionally retain older frozen coefficients as regression tripwires, so raw coefficients in those fixtures need not equal the current table.

Table 4: Claims, production modules and tests.

Claim	Mathematical anchor	Code and test anchors
Positive density, parity, monotone and convex benchmark calls	proposition 1	`backend/volfit/models/lqd/quadrature.py` `backend/tests/test_lqd_pricing.py`
Legendre basis, endpoint scales and Lee maps	(13), (20)–(28)	`backend/volfit/models/lqd/basis.py` `backend/tests/test_lqd_basis.py`

Logit quadrature, tail corrections and Hermite pricing	(15), (29)–(34)	`backend/volfit/models/lqd/quadrature.py`
		`backend/volfit/models/lqd/interp.py`
		`backend/tests/test_lqd_pricing.py`

ATM level, skew and curvature	(41)–(43)	`backend/volfit/models/lqd/atm.py` `backend/tests/test_lqd_atm.py`
Local orthogonal chart and exact retarget	(45)–(46)	`backend/volfit/models/lqd/ortho.py` `backend/tests/test_lqd_ortho.py`
Core objective, barrier, residual ordering and final rebuild	(48)–(49)	`backend/volfit/models/lqd/calibrate.py` `backend/tests/test_lqd_calibrate.py`

Claim	Mathematical anchor	Code and test anchors
Analytic Jacobian and fit agreement	theorem 1	`backend/volfit/models/lqd/jacobian.py` `backend/tests/test_lqd_jacobian.py`
Soft nearest-to-farthest calendar control	(59)	`backend/volfit/calib/calendar.py` `backend/volfit/calib/surface.py`
Active-time annualization and normalized quote preparation	(1)–(3)	`backend/volfit/api/quotes.py` `backend/volfit/api/service.py`
Fresh SVI and double-hat approximation cases	sections 10 and 11	`Docs/notes/figures/gen_lqd.py` `Docs/notes/figures/lqd_numbers.json` `backend/tests/benchmarks.py`

A Hyperparameter atlas

Table 5 lists the controls native to LQD. Shared quote, band, calendar, var-swap, prior and filtering controls are derived in Notes 07, 08, 10, 13 and 15 and indexed in Note 00 rather than duplicated here. Application defaults are distinguished from low-level function defaults where they differ.

Table 5: LQD-native controls and constants.

Knob	Default	Role
Surfaced in FitSettings		
`nOrder` N	6	Highest Legendre degree; $N+1$ parameters. Schema range $4 \le N \le 16$.
`regLambda` λ	10^{-6}	Application default for the high-order ridge. The direct `calibrate_slice` function defaults to zero.
`regPower` r	1.0	Ridge power in $\lambda n^{2r} a_n^2$ for $n \ge 4$.
`barrierCenter` c	0.90	Centre of the A_R softplus barrier; it turns the fit before the hard wall.
`barrierScale` s_b	50.0	Barrier steepness.
`weightScheme`	`equal`	Quote weights; the alternative `tv_density` scheme is described in Note 07.
Hidden numerical constants		
`Z_MAX`	40.0	Half-width of the LQD *logit integration* grid. This is unrelated to the separate quote screen at four ATM standard deviations.
`N_POINTS`	8001	Full reporting and diagnostic grid; odd so $z=0$ is a node.
`OPT_N_POINTS`	2001	Coarse grid used during optimization; accepted parameters are rebuilt at 8001.
`EPS_AR`	10^{-6}	Buffer in the code bound $A_R < 1 -$ `EPS_AR`.
`_VEGA_FLOOR` η	10^{-4}	Floor in vega-normalized quote residuals.

Knob	Default	Role
Hermite–Newton steps	4	Fixed safeguarded iterations after the linear strike-inversion seed.
`xtol/ftol/gtol`	10^{-10}	Trust-region termination tolerances.
`max_nfev`	4000	Residual-evaluation cap.

B Performance and numerical notes

1. **Cache parameter-independent arrays.** The static logit grid, logistic factors and Legendre matrix are keyed by grid size and order. This removed roughly 40% of a repeated slice build and made the benchmark build about 1.7× faster.

2. **Optimize coarse, report fine.** The 2001-node grid is already accurate well below the quote-fit budget for the supplied cases. Only the accepted parameters pay for the 8001-node rebuild. This is an empirical error budget, not a theorem for every wild coefficient vector.

3. **Store derivatives with values.** Exact nodal Q_z and $\mathcal{A}_z$ provide cubic-Hermite pricing, smoother off-grid Greeks and the interpolation data reused by the analytic Jacobian.

4. **Differentiate the cumulative objects.** Theorem 1 eliminates the strike-root derivative. The measured 1.34–1.71× optimizer-core speed-up in fig. 6 compares warm cached runs with only analytic-supported residual blocks active.

5. **Stop below economic resolution.** Trust-region tolerances of 10^{-10} avoid iterations spent chasing numerical changes far below a volatility basis point. Tightening them to 10^{-15} did not change the displayed surface in the benchmark.

B.1 Numerical qualifications

- The Black inversion accepts prices inside strict static bounds and brackets total variance over a finite interval. A failure returns a non-finite diagnostic rather than an arbitrage-free volatility by fiat.

- `martingale_check` repeats the same trapezoid-plus-tail approximation used to compute μ. It is a valuable regression self-consistency check, not an independent truncation validation.

- Stable `logaddexp` protects the softplus itself, but forming $A_R = \exp(g(1))$ can overflow for an astronomically large trial vector. The production input pipeline and trust region keep ordinary fits far from that regime.

- The low-level calibration core assumes sorted aligned strikes, positive times and variances, non-negative weights and a feasible initializer. Quote preparation enforces those conditions before the core is called.

B.2 SVI-JW provenance of the first case

The raw-SVI tuple in section 10 is the image, at $\tau = 0.5$, of the SVI-JW tuple

$$(v, \psi, p, c, \widetilde{v}) = (0.0425,\ -0.25,\ 0.75,\ 0.25,\ 0.034).$$

For completeness, with $w_0 = v\tau$,

$$b = \frac{\sqrt{w_0}}{2}(p+c), \qquad \rho = \frac{c-p}{c+p}, \qquad \chi = \rho - \frac{4\psi}{p+c}. \tag{61}$$

When $p + c > 0$, $|\chi| < 1$ and the denominator below is nonzero,

$$\sigma_{\text{SVI}} = \frac{w_0 - \widetilde{v}\tau}{b\left((1-\rho\chi)/\sqrt{1-\chi^2} - \sqrt{1-\rho^2}\right)}, \tag{62}$$

$$m = \frac{\chi\sigma_{\text{SVI}}}{\sqrt{1-\chi^2}}, \qquad a = \widetilde{v}\tau - b\sigma_{\text{SVI}}\sqrt{1-\rho^2}. \tag{63}$$

This conversion only defines the synthetic target; no SVI property is used by the LQD construction.

C Reference implementation

The following NumPy/SciPy reference keeps the mathematics visible. On the fresh $N = 6$ benchmark it reproduces the production quantile, asset share and martingale shift to approximately 10^{-8} on the dense grid; its linear off-grid interpolation is intentionally simpler than production's Hermite interpolation.

Listing 2: A compact LQD slice and call pricer.

```
import numpy as np
from scipy.integrate import cumulative_simpson
from scipy.special import expit

def legendre(nmax, x):
    P = np.empty((nmax + 1, x.size))
    P[0] = 1.0
    P[1] = x
    for n in range(1, nmax):
        P[n + 1] = ((2*n + 1)*x*P[n] - n*P[n - 1]) / (n + 1)
    return P

def build(theta, z):
    L, R, a = theta[0], theta[1], theta[2:]
    u = expit(z)
    P = legendre(len(a) + 1, 1 - 2*u)
    g = (1-u)*L + u*R + a @ P[2:]
    A_L = np.exp(L + a.sum())
    signs = (-1.0) ** np.arange(2, len(a) + 2)
    A_R = np.exp(R + signs @ a)
    if A_R >= 1:
        raise ValueError("infinite forward")
```

```python

    dz, Z = z[1] - z[0], z[-1]
    Qbar = cumulative_simpson(np.exp(g), dx=dz, initial=0.0)
    Qbar -= Qbar[len(z)//2]
    mass = np.exp(Qbar) * u * (1-u)
    total = (np.trapezoid(mass, z)
             + np.exp(Qbar[-1]-Z)/(1-A_R)
             + np.exp(Qbar[0]-Z)/(1+A_L))
    mu = -np.log(total)
    Q = Qbar + mu

    asset_mass = np.exp(Q) * u * (1-u)
    A = cumulative_simpson(
        asset_mass[::-1], dx=dz, initial=0.0
    )[::-1]
    A += np.exp(Q[-1]-Z)/(1-A_R)
    return u, Q, A, mu

def call(k, z, u, Q, A):
    zk = np.interp(k, Q, z)
    return np.interp(zk, z, A) - np.exp(k)*(1-expit(zk))
```

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